

Programming Using C++ Level 1

Do While Loops



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Definition

◆ A do-while loop is a control structure that executes a block of code at least once and then repeats it while a condition is true.

Explanation

- ◆ A while loop checks the condition before executing the code.
- ◆ A do-while loop executes the code first, then checks the condition. Therefore, the code inside a do-while loop always runs at least once.
- ◆ In this program, the user must enter a number between min and max. If the number is outside the range, the program asks again.

Steps in main()

1. Set the minimum and maximum values.
2. Ask the user to enter a number.
3. Store the number in Number.
4. Check whether Number is outside the range.
5. If it is outside the range, ask the user again.
6. Repeat until the number is within the valid range.
7. Return the valid number.

💡 Example

For ReadNumberInRangeDo(18, 45)

If the user enters 10:

Wrong value → asks again.

If the user enters 50:

Wrong value → asks again.

If the user enters 30:

The value is valid → returns 30.

➡ Result

The program keeps asking for a number until the user enters a value between 18 and 45. and 20, the program displays:

The sum is: 30

Note

- ◆ The do-while syntax ends with a semicolon after the condition:

```
do
{
    // code
}
while (condition);
```


⊘ Warning

- ◆ Do not forget the semicolon after the while condition in a do-while loop.



Comparison

▨ while loop

- ✓ Checks the condition before executing the loop body.
- ✓ The loop body may execute zero times.
- ✓ In this example, the first input is read before the while loop.

▤ do-while loop

- ✓ Executes the loop body before checking the condition.
- ✓ The loop body always executes at least once.
- ✓ Useful when the user must provide input at least once.

✦✦ Summary

- ✓ A do-while loop executes its code at least once.
- ✓ The condition is checked after each iteration.
- ✓ It is useful for input validation when the user must enter a value at least once.

Thank you 😊